The title

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Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline. Two to three sentences of **more detailed background**, comprehensible to scientists in related disciplines. One sentence clearly stating the **general problem** being addressed by this particular study. One sentence summarizing the main result (with the words "**here we show**" or their equivalent). Two or three sentences explaining what the **main result** reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge. One or two sentences to put the results into a more **general context**. Two or three sentences to provide a **broader perspective**, readily comprehensible to a scientist in any discipline.

Keywords: keywords

Word count: X

The title

Death is something each of us must learn to cope with, whether in healthy ways or less so. These issues may be at front of mind for many in light of the COVID-19 pandemic. Various existential philosophers and psychologists have proposed ways in which we deal with the awareness of death and the anxiety this awareness often causes. Psychoanalyst Erik Erikson (1950) proposed that during mid-life one becomes acutely aware of their oncoming death and is motivated to care for things which will outlast themselves. He called this act of caring generativity. In *The Denial of Death*, philosopher Ernest Becker (1973) posits that humans undertake immortality projects to curb their sense of vulnerability to death. Similarly, psychiatrist Robert Jay Lifton (1979), a mentee of Erikson, described the awareness of death as being ever present and motivating us to create symbols, thereby allowing us to imagine ourselves as symbolically immortalized. Existential psychiatrist Irvin Yalom (2008) notes that many of his clients experiencing anxiety about their death take comfort in "rippling," the idea that one's lasting effects on the world will ripple out and influence the world after they have died.

Although these thinkers use different terminology, there are several common themes among their ideas. (1) Our physical death is an inevitability, and we often find our awareness of its inevitability to be aversive. This aversion may be referred to variously as angst, death-anxiety, despair, being-towards-death, terror, and so on. However, (2) we take comfort in the idea that other, non-physical parts of us continue to exist indefinitely after our biological death, through mechanisms such as the heroic archetype and symbolic self. (3) Finally, we can take action to promote these non-physical aspects of the self, such as through search for meaning, sense of immortality, care, generativity, and rippling.

One of these bodies of thought, called symbolic immortality, was originally theorized by Lifton (1979), who thought that awareness of death drives a fundamental human desire for a sense of continuity lasting beyond the lifespan. Essentially, humans are

meaning-seeking creatures, and throughout our lives, this search for meaning involves an evolving psychological imagery of life and death. Death, or the transient nature of life, threatens our search for meaning. Lifton thought that if we could achieve what we believe to be some form of immortality, we could overcome this loss of meaning, and the awareness of death could instead drive an inner vitality (imagery associated with connection, integrity, and movement). If this drive toward vitality is lost, we are vulnerable to a psychic numbness or death-in-life (imagery associated with separation, disintegration, and stasis). In Lifton's own words, "Death does indeed bring about biological and psychic annihilation. But life includes symbolic perceptions of connections that precede and outlast that annihilation" (1979, p. 18).

Lifton (1979) proposed five modes of experience or ways of achieving symbolic immortality: The biological (or biosocial) mode in which one lives on through their genetic and sociocultural progeny, the creative mode in which one's accomplishments and contribution outlast oneself, the natural mode in which one feels they are a part of the broader universe, the spiritual mode in which one seeks to transcend the physical realm to a higher spiritual realm beyond death, and the mode of experiential transcendence in which one experiences a phenomenological state of flow. The experiential mode must occur in the context of at least one of the other four to really be considered transcendent, but it is thought to have a great capacity to bring about personal change.

Claims of how we suppress death-anxiety have been investigated experimentally, primarily through the paradigm of Terror Management Theory (TMT). Based on the theories of Ernest Becker, TMT posits that human awareness of death is always present to some degree. This awareness of our inevitable death, coupled with a strong aversion to thoughts of death, causes terror and is pushed out of our consciousness by our creation of meaning systems (Greenberg, Pyszczynski, & Solomon, 1986). TMT proposes that self-esteem, interpersonal relationships, and cultural worldview work together to buffer

against our anxiety about death. It is assumed that these buffers suppress thoughts of death by providing a sense of symbolic immortality, though little systematic research has been conducted on this construct. The results of this buffering process are not always positive. For example, experimentally priming mortality salience can lead to more positive attitudes toward in-group members but harsher negative attitudes toward out-group members (Greenberg et al., 1990).

TMT refers to a person's awareness of death as mortality salience (MS). The MS hypothesis of TMT posits than an increase in one's awareness of death causes an increase in compensatory behaviors to lower their death anxiety, either by distracting from the awareness of death or by the promotion of meaningful cultural worldviews. In the MS paradigm, experimentally priming a participant's awareness of death (for example, by having participants write about death and then complete a distraction task) is thought to cause an increase in compensatory buffers. A meta-analysis of 277 experiments found mortality salience to have a robust, moderate overall effect size: $r(276) = 0.35$, $p = .00$ (Burke, Martens, & Faucher, 2010). Altogether, these experiments provide convincing evidence for TMT and the MS hypothesis in particular.

Though some avoidance of (or buffering against) death anxiety is thought to be universal and has the potential to increase interpersonal conflict, awareness of death through symbolic immortality may also have potential as a positive force. In particular, it is thought to be an underlying motive for what Erikson referred to as generativity. Generativity is the seventh of eight proposed stages in Erikson's (1950) theory of psychosocial development, which he associated with midlife and described as "the concern in establishing and guiding the next generation" (E. H. Erikson, 1963, p. 267). Little systematic research was conducted on this subject until the 1980's. Kotre (1984) expanded on the theory and proposed that the drive for generativity was related to a motive to expand the sense of self beyond the lifetime, especially in light of the fear of death.

McAdams and de St. Aubin (1992) sought to formalize the study of generativity as a multidimensional construct. Their seven components of generativity include cultural demand, inner desire (for symbolic immortality and community), concern (for the next generation), belief (in the human species), commitment, action, and narration (of generativity within one's life story). In addition to a quantitative measure of generative concern (the Loyola Generativity Scale), they developed a system for content analysis of autobiographical episodes pertaining to generativity, and symbolic immortality is one of the five themes they found. Here they define symbolic immortality as "any reference to leaving a legacy, having an enduring influence, or leaving behind products that will outlive one's physical existence," a theme clearly related to both Lifton's and Erikson's theories (McAdams & St. Aubin, 1992, p. 1011).

These research areas depend on the construct of symbolic immortality for their theoretical frameworks, but few researchers have attempted to systematically and quantitatively assess this construct. Two attempts have been made to develop such a measurement: Drolet's (1990) Sense of Symbolic Immortality Scale and Mathews and Kling's (1988) measure of symbolic immortality, based on an original questionnaire by Mathews and Mister (1987).

Drolet (1990) developed the Sense of Symbolic Immortality Scale based on Robert J. Lifton's theory of symbolic immortality and its five modes of experience. Drolet studied 136 adults, ages 18-30 and 30-40, and hypothesized that those in their 30's (established adults) would have a greater sense of symbolic immortality than the young adults (18-30). The measure is inherently subjective, not only by the nature of self-report, but in that the scale seeks to measure what a person *believes* and how they *feel* about these subjects. The scale as a whole had a high internal consistency ($\alpha = .91$) and test-retest reliability was $r = .97$. Internal consistency of subscales for the five theoretical modes of immortality was mixed. Of the five, spiritual immortality was the most distinct from the scale as a whole and the other

subscales. Factor analysis showed three factors, mapping onto biosocial, creative, and spiritual. The transcendent and natural items may be closely related to biosocial.

Moving beyond the scale development itself, SSI correlated negatively with Templer's Death Anxiety Scale and had a strong ($r = .84$) positive relationship with Maholick's Purpose in Life Test (Drolet, 1990). In interpreting the very strong correlation, the author suggests that SSI is a broader construct than Purpose in Life and the scale itself may be less prone to social desirability effects than the PIL, although this had not been directly tested. Age group was also related, with established adults having a higher SSI, particularly in the biosocial and creative domains.

We see multiple issues with using the Symbolic Immortality Scale. First, the study was underpowered, conducting exploratory factor analysis of 67 items using a sample of 136. Second, the scale was developed in French, and we do not take for granted the psychometric properties of a translated version. Third and most fundamentally, the scale has poor face validity and appears to measure the constructs theorized to symbolically immortalize rather than a sense of symbolic immortality directly. For example, the scale includes items such as "My sex life contributes greatly to my well-being", "Intimate relationships scare me", and "I am sure of who I am." Although related to the constructs (such as interpersonal relationships and self-esteem) which theoretically help cope with death, it is unclear how these items represent the construct of symbolic immortality itself.

Mathews and Mister (1987) also developed a scale pertaining to symbolic immortality, sensation seeking, and psychic numbness in a study including 400 adults. Experiential transcendence was operationalized as similar to Zuckerman's (1979) sensation seeking, which may not fully capture the original intent (the experience of losing oneself). Items were mapped onto five factors, and the five factors largely aligned with Lifton's constructs. Although internal consistency was at least acceptable for each factor, goodness of fit statistics are not reported. Some studies have used a revised version of the scale by Mathews and

Kling (1988), who adapted it for a study on prosocial behavior in the context of nonprofit volunteer motivation. They reported similar results for their revised scale. The items on these scales seem to have more face validity than the scale by Drolet, but some factors seem more behavioral and unnecessarily specific: pertaining to one's religiosity or biological children, whereas Lifton's theory allows for a broader interpretation of these dimensions. The Nature and Creative factors seem most useful and theoretically aligned with Lifton.

Much more advanced factor analysis methods have been developed since the 1980s, but to our knowledge, these scales have not been tested with more robust tools. The goal of the present research is to develop an up-to-date symbolic immortality scale that more directly measures one's sense of symbolic immortality and which contains items more generally applicable to broad groups of participants (e.g., regardless of a person's religious beliefs and parental status). We have attempted to use current best practices for scale development and analysis.

Method

Participants

Participants were recruited from two data sources. 564 undergraduate students taking introduction to psychology were recruited for course credit at a large midwestern university. Undergraduate participants were $M = 19.62$ ($SD = 2.75$) years old at the time of the study. 240 participants were recruited from Amazon's Mechanical Turk, and these participants were $M = 35.55$ ($SD = 11.05$) years old. No other demographic information was collected in the study.

Material and Procedure

55 items were developed using previous scale development on generativity (Drolet, 1990), along with theoretical factor alignment to XXCITEXX. These items aligned with areas of biological, physical, social, spiritual, and personal immortality. The survey was

implemented in Qualtrics. First, participants entered their age in years for basic demographic information. The immortality items were separated into three blocks of 20-21 questions. These three blocks of questions were randomized across participants, and the items within each block were also randomized. In each block, one attention check item was shown (e.g., "Answer this question as somewhat agree."). At the end of the study, participants were directed back to either SONA for undergraduate course credit or Mechanical Turk for monetary compensation[1].

Data analysis

This study was not pre-registered. We conducted data analysis in two phases. The data were first screened for inattentive and outlier responses. The data were then randomly split into an exploratory and confirmatory dataset. We developed the XX scale by using exploratory factor analysis (EFA) with maximum likelihood estimation and an oblique (i.e., oblimin) rotation (Preacher & MacCallum, 2003). Analyses were performed with the *psych* package (Revelle, 2020). We used scree plots, parallel analysis, and theory to determine the number of factors and compared models based on fit indices to select the best model. We use the following fit indices: Root Mean Squared Error of Approximation (RMSEA, Steiger, 2016), Root Mean Squared Residual (RMSR, Jöreskog & Sörbom, 1996), Comparative Fit Index (CFI, Bentler, 1990), and the Tucker-Lewis Index (TLI, Tucker & Lewis, 1973). Values close to one are expected for goodness of fit statistics (CFI, TLI, Bentler, 1990; Hu & Bentler, 1999), while lower values close to zero are considered better models for residual statistics (RMSEA, RMSR, Steiger, 2016).

We then analyzed the best model from the exploratory stage using confirmatory factor analysis (CFA) and multigroup CFA (Brown, 2015; Kline, 2016) on the second confirmatory dataset. The models were analyzed with *lavaan* (Rosseel, 2012) and *visualizemi* (Buchanan,

[1] Ûnfortunately, the exact amount we paid participants is lost to time. Given our ethics application, we believe it was 1-2 USD, and the survey was completed in $M = 6.34$ ($SD = 4.85$ minutes.

2024) *R* packages. Multigroup CFA can be used to determine if the scale structure can be generalized to subsamples within the data, and thus, allowing for the same interpretation of total or subscale scores calculated from the scale. First, we modeled the overall structure from the EFA results to determine fit in a new sample. The RMSEA, CFI, and TLI were used to determine model fit along with the Standardized Root Mean Squared Residual [SRMR; Bentler (1995)]. The same interpretations were included from the exploratory stage: high fit indices close to one for goodness of fit statistics, and low fit indices close to zero for residual fit statistics. If the overall model fit well, we then modeled the subsamples of student populations and Mechanical Turk populations separately. While fit is expected to potentially degrade due to smaller sample sizes, we expect these models to converge to demonstrate that it is appropriate to combine models for multigroup testing.

For the multigroup CFA, we used Brown (2015)'s suggested terminology and steps for analysis. The *configural* model includes both subsamples combined together into one model controlling for group structure. This model supports the "configuration" or structure of the scale is at least similar within each group, but does not force each group to show similar patterns of parameter estimates. Each subsequent step constrains a set of parameters to be equal across groups to demonstrate invariance (i.e., equivalence). The configural model is then compared to the *metric* model, which sets the factor loadings for each measured item to be equal across groups. This model supports that each item is related to the underlying latent variable the same way for each group. If this assumption holds, the metric model is then compared to a *scalar* model, which sets the item intercepts (i.e., item means) to be equal across groups. Scalar model invariance suggests that each subsample shows the same item means and factor loadings across groups. Finally, we compared the scalar model to the *strict* model, which constrains item residuals (i.e., errors, variance) to be the same across groups. Invariance in model residuals suggests that the items have the same answer range/spread across groups.

To assess if models are invariant, we used Δ CFI <= .01 criterion to indicate that models were equivalent (Cheung & Rensvold, 2002). If models were not considered invariant, we investigated for the source of the invariance by using partial invariance testing (Byrne, Shavelson, & Muthén, 1989; Cheung & Rensvold, 1999). Each parameter estimate is individually relaxed to determine if that particular estimate leads to misfit in the model. For example, if the model fails scalar invariance, each item mean would be allowed to freely vary between groups (one at a time) to find the items that show group differences. If only a few items show group differences, then the model is considered "partially" invariant, and the impact on total scores can be assessed. Partially invariant models indicate areas for further investigation into group differences and ways that the final scores of the model must be carefully interpreted. Previously, Nye and Drasgow (2011) and Nye, Bradburn, Olenick, Bialko, and Drasgow (2019) proposed an effect size d_{MACS} to help quantify the size of the difference between groups in partially invariant models. However, d_{MACS} can generally only be used to understand the item means and loadings together (due to the mathematical calculation), regardless of the tyep of non-invariance. Therefore, we applied a new technique proposed by Buchanan (2024) to assess several potential effect size components of the model: 1) overall model replication likelihood, 2) individual parameter replication likelihood, and 3) effect size for individual parameters showing non-invariance.

We will use the h_{nmi} effect size as the likelihood of replication statistic for both the overall model and individual parameters. First, the data is bootstrapped with replacement n times to create similar samples to the original data. The grouping variable is then randomized to create a comparison model. The proportion of times that invariance (or non-invariance) is met is calculated for the bootstrapped and randomized group labels. h_{nmi} is Cohen's h , which is the standardized difference between two proportions for non-invariance (no measurement invariance, but the statistic can also be calculated as h_{mi} for the difference in proportions for when the model showed measurement invariance). We suggest h_{nmi} because the interpretation is a bit simpler: effect sizes close to zero indicate

that the bootstrapped data and randomized groups data show the same number of non-measurement invariance findings, indicating that they are likely the same result. The logic here is that if groups are truly equal in their measurement on a scale, then the grouping variable should not affect the results, thus, randomizing the grouping variable label does not affect the results. As h_{nmi} increases, we expect to find non-measurement invariance between groups, as the bootstrapped data is consistently different than randomized group labels. h can be interpreted much like d, but d does not technically have upper or lower limits (i.e., they are ∞), while h's range is limited to π. Therefore, we will report h_{nmi_p} which is scaled to range from -1 to +1 for easier interpretation. In addition to this effect size, we can calculate d for the differences in parameters for bootstrapped versus random group labels, thus, providing an estimate for the size of group differences on partially invariant parameter estimates.

Results

Data Screening

Undergraduate Data. Data were screened for accuracy, missing data, outliers, and multivariate parametric assumptions. The undergraduate data included $n = 564$ participants. We first screened each participant for missing data and excluded $n = 7$ participants for missing more than 5% of their data (Tabachnick & Fidell, 2013). 105 participants included less than 5% missing data (i.e., 3 items or less), and these data points amounted to less than 2% of data missing per question. These items were replaced with a random imputation from the *mice* library (Buuren & Groothuis-Oudshoorn, 2011).

The data were then screen for attentiveness and accuracy using the framework from Buchanan and Scofield (2018) to find low quality responses. The survey items were split into three pages, and each page was screened separately using click counts, an attention check item (i.e., please answer strongly agree for this item), and number of characters on a page for reading speed. The total number of indicators for low quality data was summed for each

participant across pages (e.g., 3 indicators by 3 pages = 9 total possible items), and $n = 49$ participants showed at least 4 indicators across all pages. Buchanan and Scofield (2018) suggest a criteria of 2/5 screeners or 40%, and therefore, we used 4 items as the ~40% cut off for inclusion in the data. The final sample size for the undergraduate population was $n = 508$. We additionally screened for multivariate outliers using Mahalanobis distance across items and found $n = 43$ possible outliers, but they are included in the analyses described below.

Mechanical Turk Data. The Mechanical Turk data were screened in the same manner as the undergraduate sample. The original sample size was $n = 240$ participants. $N = 23$ participants were excluded for missing more than 5% of their data, and $n = 36$ participants had 3 or less data points estimated using the *mice* library. The same low quality data screening was performed, and $n = 20$ participants were excluded for having 40% or more low quality indicators. Last, $n = 10$ participants were flagged as potential outliers but were not included in analyses presented below.

The data was then randomly split into $n = 352$ participants for the exploratory factor analysis, and $n = 353$ for the confirmatory factor analysis. The EFA data included n undergraduate = 254 participants, and n Mechanical Turk = 98 participants. The CFA data included n undergraduate = 254 participants, and n Mechanical Turk = 99 participants.

EFA

Number of Factors. To determine the number of factors that best fit the data, we used suggested theory (5 factors), a scree plot, and a parallel analysis using the *psych* library (Revelle, 2017). Parallel analysis suggested 7 factors, while a visual inspection of the scree plot indicated 2 to 3 factors. We analyzed a 2-factor, 3-factor, and 5-factor model for simple structure and fit indices.

Simple Structure. The 58 items were first tested with a 2-factor fit, and 7 items were excluded for not loading on any factor or loading on multiple factors with a loading greater than .30. After removing these items, all items loaded cleanly on the two factors.

The loading table for this analysis can be found on our GitHub repository (links provided in the appendix). This same procedure was repeated for the 3-factor fit, and 10 items were removed from the first round of analysis, and 1 item was removed from the second round of analysis, resulting in a final set of 47 items. The loading table and items can be found in the online supplemental materials. Last, the 5-factor model showed 15 poor items on the first round of analysis, and 10 poor items on the second round of analysis, resulting in a final set of 33 items for the final model. The items and loadings can be found in the online supplemental materials.

We then compared items across all factor models to reduce scale length. Items were selected for the final model if they: 1) theoretically made sense with the other items on that factor, 2) loaded on the theorized 5-factor model (i.e., items were included in the 2-factor model if they also had been included in the 5-factor model), and 3) were not face-value redundant with other items on that factor. We tested these items and found that the 2-factor and 3-factor models continued to show simple structure, but the 5-factor model required the removal of 2 more items due to cross loadings across factors. To select the best model, we examined the fit indices for each factor structure. The 5-factor model showed the best fit (RMSEA = .038, RMSR = .024, TLI = .968, CFI = .982), followed by the 3-factor model (RMSEA = .049, RMSR = .028, TLI = .962, CFI = .977), and then the 2-factor model (RMSEA = .062, RMSR = .031, TLI = .963, CFI = .977). The final factor loadings for the 5-factor scale are shown in Table 1, and the other loadings can be found in the online supplemental materials.

```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘ML1 = apa_num(ML1, digits = 3, gt1 = F)‘.
## Caused by warning in ‘apa_num.numeric()‘:
## ! You specified gt1 = FALSE, but passed absolute value(s) that exceed 1.
```

EXPLAIN ITEM LABELS HERE

CFA

Using the multigroup CFA procedure described above, we examined the 5-factor model for overall fit and fit across groups. Table 2 portrays the results of the assessment of all models. The model showed acceptable fit overall with lower than desired values on the TLI, and lower values on the CFI and TLI for each individual group. Model fit appeared acceptable to investigate multi-group structure when examining the configural model with good values for the residual fit statistics, and poorer values for the goodness of fit statistics. We examined for potential areas of misfit using modification indices but did not find any suggested correlated error terms (i.e., two similar items that have related residuals) or cross-loadings that fit within theoretical structure. Therefore, we did not make any modifications.

Partial Scalar Invariance. Using the ΔCFI $\leq$ .01 rule for degradation in fit, we found that the loadings (*metric*) showed invariance across groups, while a large degradation in fit was found between the loadings and intercepts (*scalar*). We individually removed the equality constraint between groups for each item, and found one item that was different between groups: *I have children or students which take after me.* The intercept for the undergraduate sample on this item was: 2.26, $SE = 0.08$, while the Mechanical Turk sample showed a higher average item score: 3.45, $SE = 0.12$. Figure 1 demonstrates a visualization of those differences with higher intercepts for the Mechanical Turk group. The overall distribution for the biological factor is also displayed in this figure.

Partial Strict Invariance. Following the partial intercepts model, we found non-invariance when all residuals were constrained to equal. The same steps were applied and 2 non-invariant items were discovered that would allow the model to be partially invariant. First, *I believe in an afterlife* showed larger residuals for the undergraduate population 0.36 ($SE = 0.05$), while the Mechanical Turk population showed a much smaller amount of variance 0.06 ($SE = 0.03$). The second item was a similar item, *My spirit or soul will live on after I physically die*, where the undergraduate population showed a much larger

variance 0.55 ($SE = 0.06$) than the Mechanical Turk population 0.24 ($SE = 0.05$). Figure 2 demonstrates the differences in variance and overall latent variable means for these items on the spiritual factor.

Model Replication and Effect Sizes. h_{nmi} represents the effect size of the model in comparison to a randomized groupings model for non-measurement invariance, and effect sizes close to zero indicate that there is no difference in the actual group labels and the randomized group labels. Thus, we would expect to find *invariance* because the group labels show the same patterns of data. The h_{nmi_p} effect size is scaled from -1 to +1 for easier interpretation. For metric invariance, we found an effect size of $h_{nmi_p} = 0.31$ indicating that some models may show non-invariance when replicated with the same groups. However, the effect size for scalar invariance showed twice the effect size, $h_{nmi_p} = 0.63$, indicating that it is likely that we would again show differences in groups for our non-invariant scalar items.

At this stage, we examined the partial invariance for both the scalar and strict models that indicated non-invariant items above. Figure 3 portrays the effect sizes (ds) for each questions when bootstrapped with the current group labels or randomized group labels. Figure 4 indicates the distribution of scores for each item in the same order. In these figures, we can see potential other items that may produce non-invariant results. Question 4_8 (*I have children or students which take after me.*) showed non-invariance in our MGCFA steps; however, multiple items show large effect size differences in item-intercepts with higher scores for the undergraduate group, even though they were found to be invariant in the model analyses. Figures 5 and 6 show the same effects for the residual variances in our bootstrapped model. Questions 4_19 (*I believe in an afterlife*) and 2_17 (*My spirit or soul will live on after I physically die*) were our previously found non-invariant items. The same patterns appear within the residuals with other potential items showing large effect size differences. A close inspection of the raw scores for figures reveals that the item intercepts and residuals may potentially be slightly bimodal. In this scenario, the one sample would show invariant results, but many similar samples may show non-invariant results depending

on the composition of the final sample groups.

Discussion

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Appendix

All materials, code, and the reproducible manuscript can be found at: https://github.com/doomlab/gen-scale.

- EFA Model Loadings: Loadings for models before they were combined and examined together.
  - Two-Factor: https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/2_factor.xlsx
  - Three-Factor: https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/3_factor.xlsx
  - Five-Factor: https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/5_factor.xlsx
- EFA Model Loadings Reduced: Loadings for models when items were compared across models and then re-analyzed.
  - Two-Factor: https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/2_factor_reduced.xlsx
  - Three-Factor: https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/3_factor_reduced.xlsx
  - Five-Factor: https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/5_factor_reduced.xlsx
- Final Questions: Final question set for those who want to use the scale.
  - https://github.com/doomlab/Gen-Scale/raw/refs/heads/master/05_Manuscript/results/5_factor_reduced.xlsx
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Table 1

EFA Loadings for the 5-Factor Model

Spiritual	Social	Personal	Biological	Physical	Item
.928	-.045	-.003	.017	-.007	I believe in an afterlife.
.907	.032	.000	-.017	.000	My spirit.or soul will live on.after I physically die.
.828	-.011	-.042	-.005	.068	I believe in an immaterial soul or spirit.
.816	.011	.041	.037	.004	Death is not the end.
.509	.146	.066	-.032	-.123	The ideologies I believe in such as my religion, philosophies or politics will live on.
.009	.790	.080	-.081	-.068	I have taken part in activities or events which will impact the future.
-.053	.758	-.004	.028	.146	I have contributed to the world in a unique way.
.023	.733	-.061	.122	-.048	I have contributed to things which will be meaningful.
.063	.645	.037	-.056	.162	My life has influenced the world.
.123	.592	.083	.101	-.170	I feel I have taken part in something bigger than myself.
.131	.347	.067	.216	.132	I believe I will be remembered for a long time after I die.
.001	-.024	1.020	-.024	.027	Parts of my personality.have rubbed off on other people.
.028	.083	.692	.115	-.019	Through my interactions with them. others have picked up parts of my personality.
-.065	.290	.347	.170	-.106	I have imparted knowledge and skills on to others.
.028	-.035	-.042	.759	.044	My offspring carry parts of me with them.
.017	.018	.081	.731	.001	I am comforted by the fact that my genes can/will be carried on in the future.
.081	.109	.139	.444	-.095	I am comforted by knowing my loved ones will live on after me.
-.053	.183	-.046	.348	.203	I have children or students which take after me.
-.012	.013	-.011	.053	.749	I believe will not die.
-.011	-.010	.102	.005	.601	I have a hard time believing I will physically die.
.129	.031	-.020	-.006	.618	I don't entirely believe I will die.
.068	.044	.072	-.018	.470	I believe I will be reincarnated.

Table 2

Model	Chi-Square	df	CFI	TLI	RMSEA	SRMR
Overall	503.453	76	0.901	0.885	0.066	0.065
Group Undergraduate	433.000	76	0.876	0.856	0.068	0.070
Group Mechanical Turk	331.121	76	0.899	0.883	0.082	0.094
Configural	764.120	152	0.885	0.867	0.072	0.077
Loadings	790.521	135	0.882	0.869	0.072	0.082
Intercepts (Original)	895.284	118	0.855	0.845	0.078	0.086
Intercepts (Partial)	830.685	119	0.875	0.866	0.072	0.085
Residuals (Partial Intercepts)	914.583	97	0.855	0.852	0.076	0.085
Residuals (Partial Intercepts + Residuals)	873.223	99	0.868	0.864	0.073	0.085

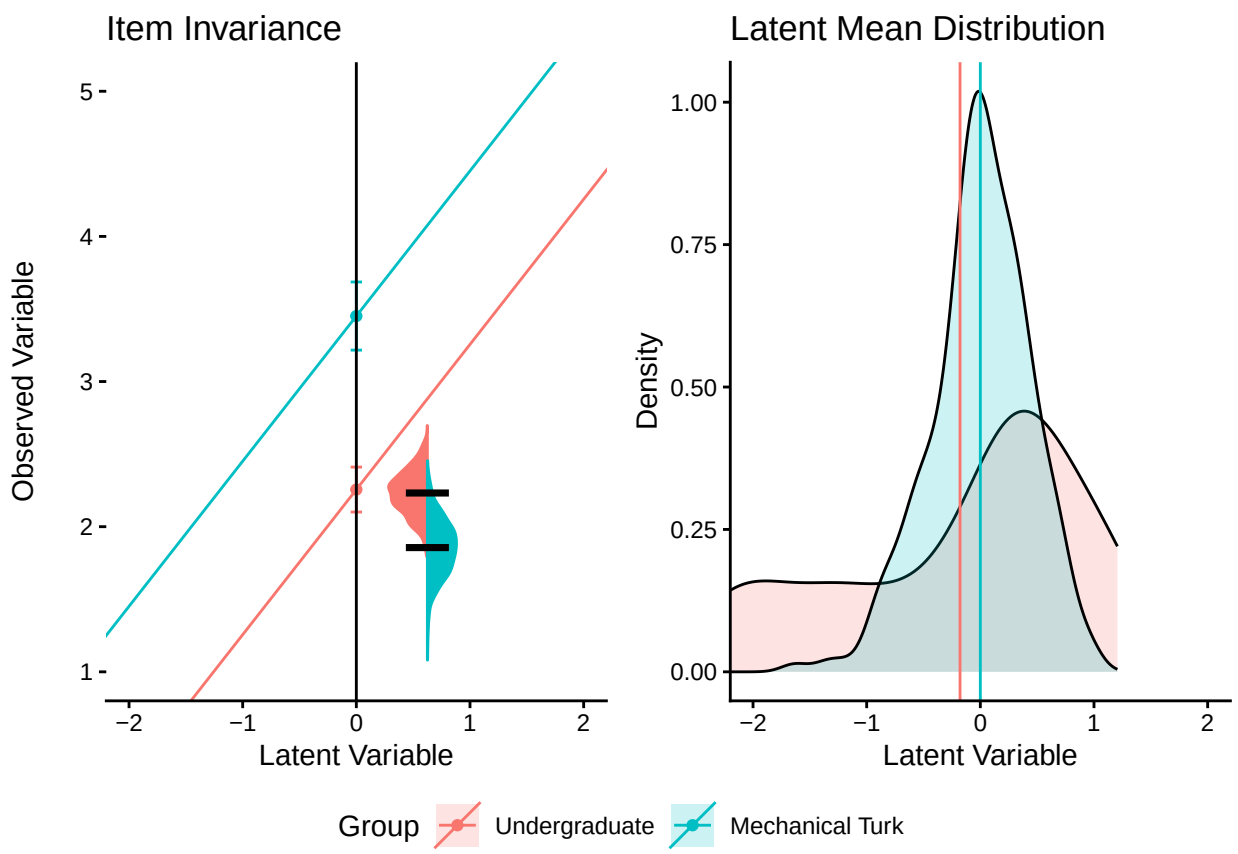

Figure 1. The left panel shows the loadings (slope of the regression lines), intercepts (where each line crosses the y-axis), and residuals (inset bottom right) for the non-invariant biological item. The loadings are set to be equal, but the large difference in item averages can be seen as the separation between lines. The right panel indicates the shape of the distribution of the latent variable (which includes all biological items). The Mechanical Turk group shows a normal distribution around the latent variable mean (set to zero as part of MGCFA procedures), and the undergraduate population shows a negatively skewed distribution.

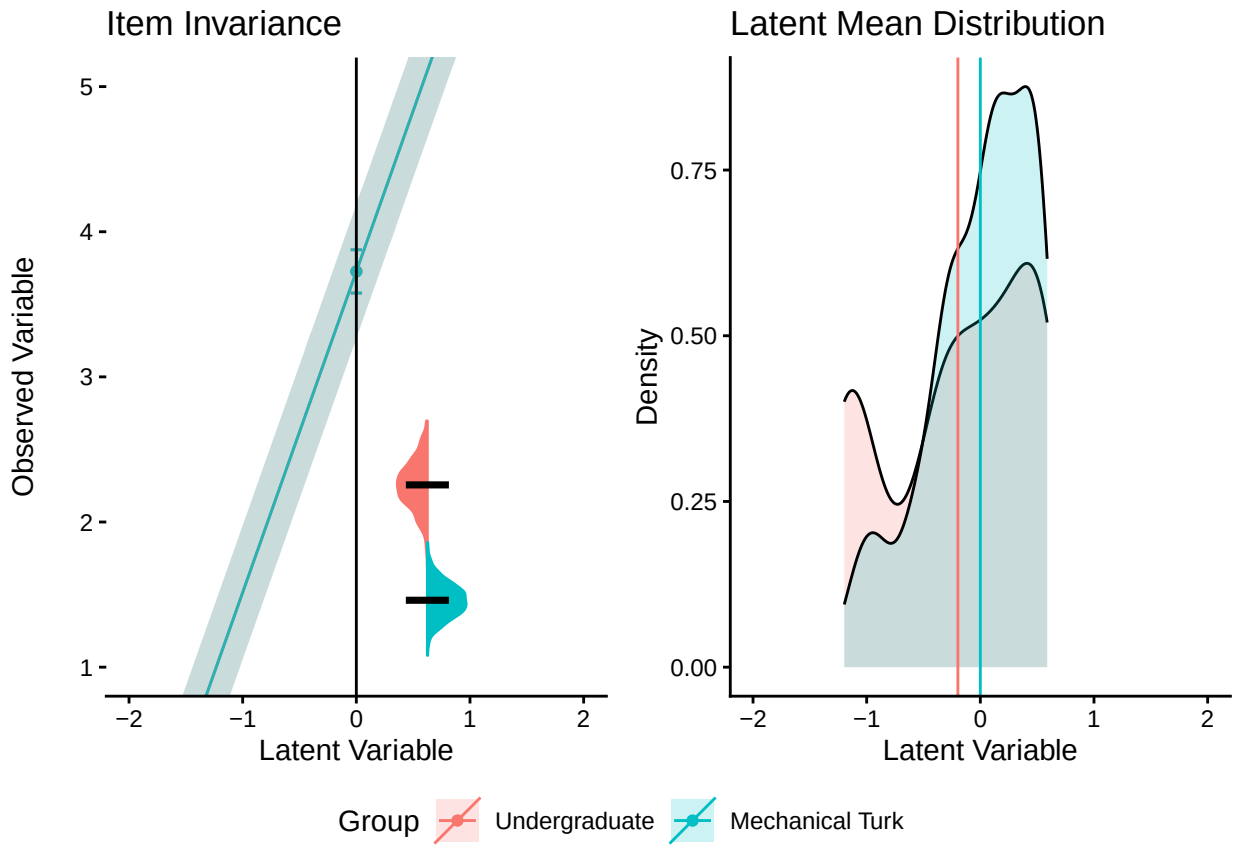

Figure 2. In the left panel, the large difference in residuals can be found for the spiritual items, as loadings and intercepts are set to be equal. The placement of residuals on the plot is unrelated to the x- or y-axis and simply shows the magnitude of the difference. Both items showed nearly identical plots, and therefore we only plotted the afterlife question. The right panel shows the latent mean distribution for the entire spiritual set of items.

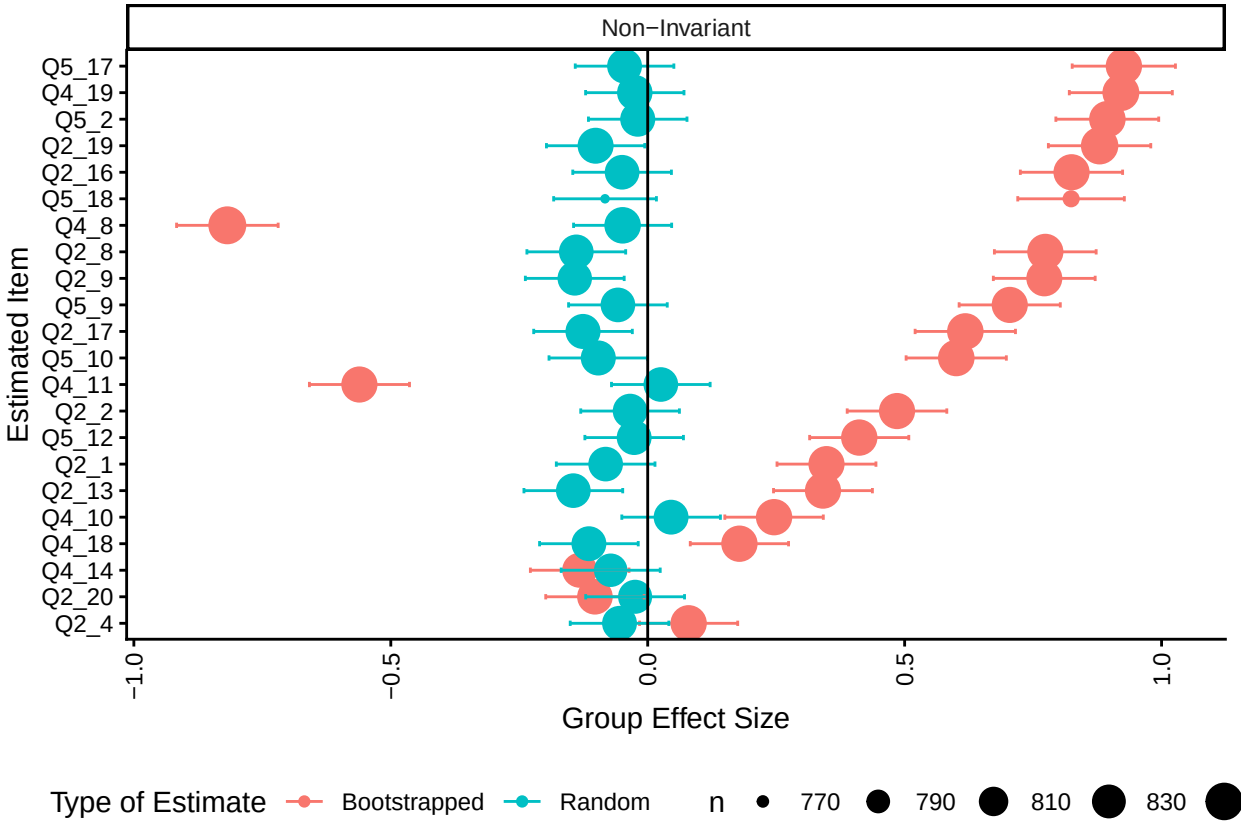

Figure 3. Effect size differences (d) between groups for item intercepts when the partially invariant scalar model is bootstrapped or randomized. Question numbers correspond to our online materials and are organized in descending magnitude. Sample size represents the number of bootstraps out of 1000 that were non-invariant and converged appropriately.

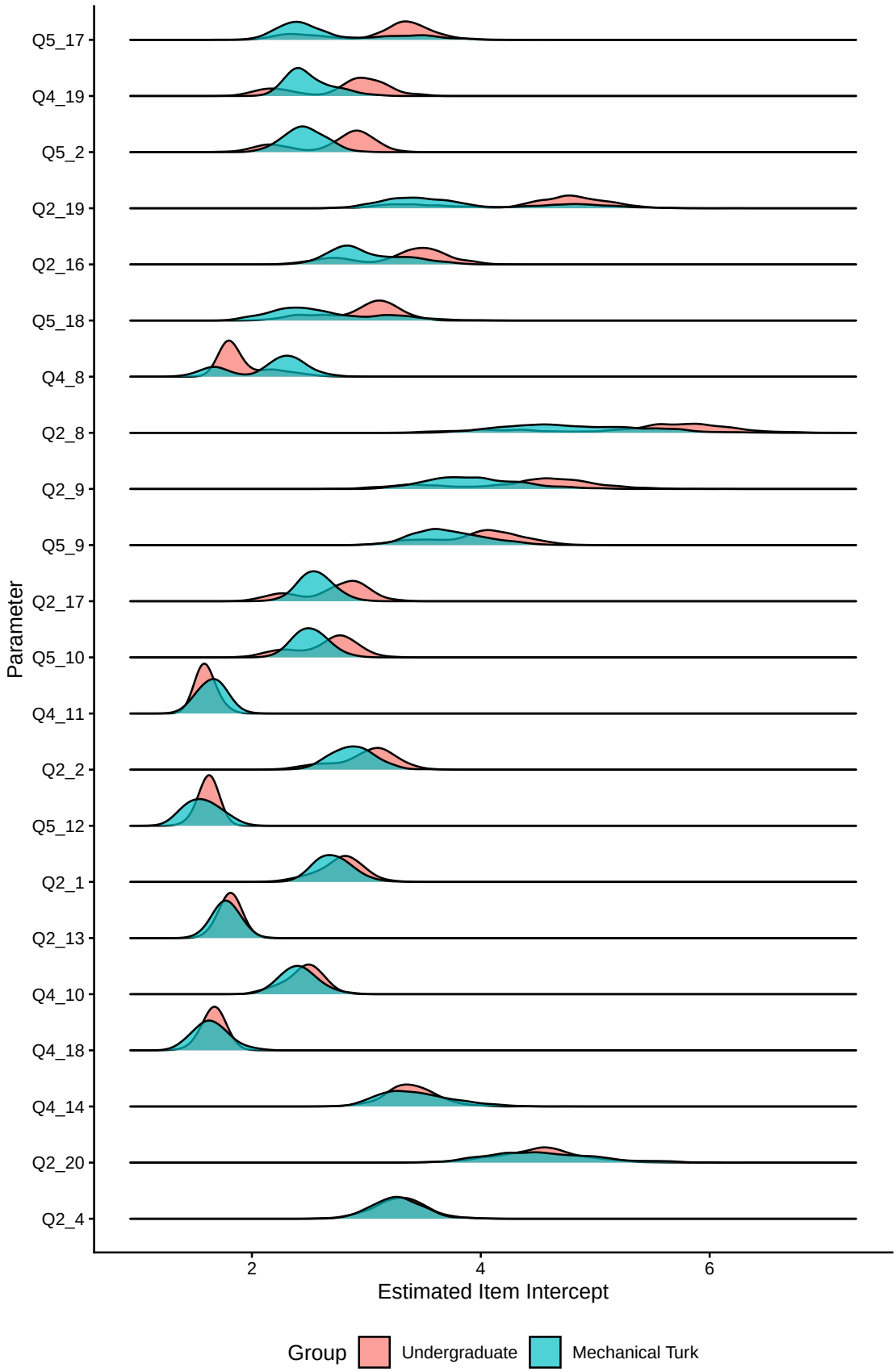

Figure 4. Item intercept distributions for items that were bootstrapped with original group labels and were non-invariant ordered by effect size of the item intercept difference.

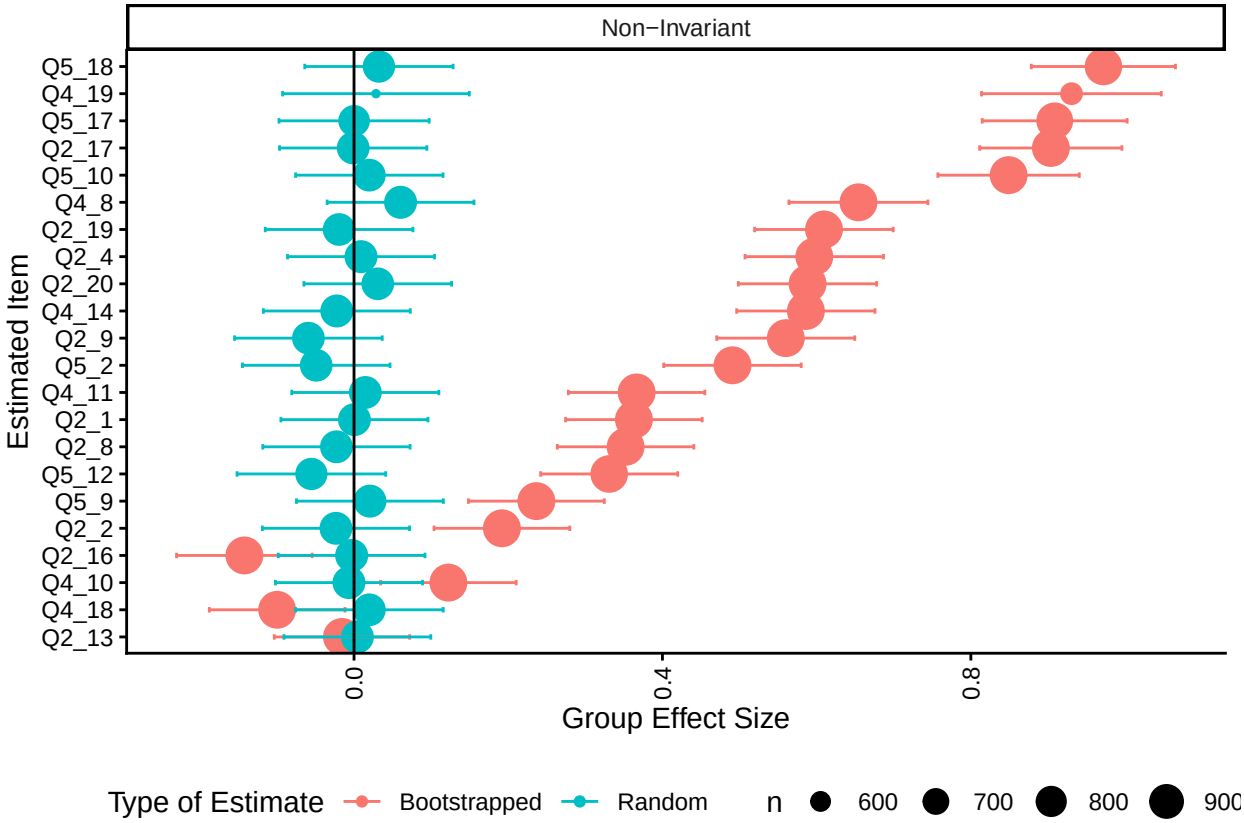

Figure 5. Effect size differences (d) between groups for residuals when the partially invariant residual model is bootstrapped or randomized. Question numbers correspond to our online materials and are organized in descending magnitude. Sample size represents the number of bootstraps out of 1000 that were non-invariant and converged appropriately.

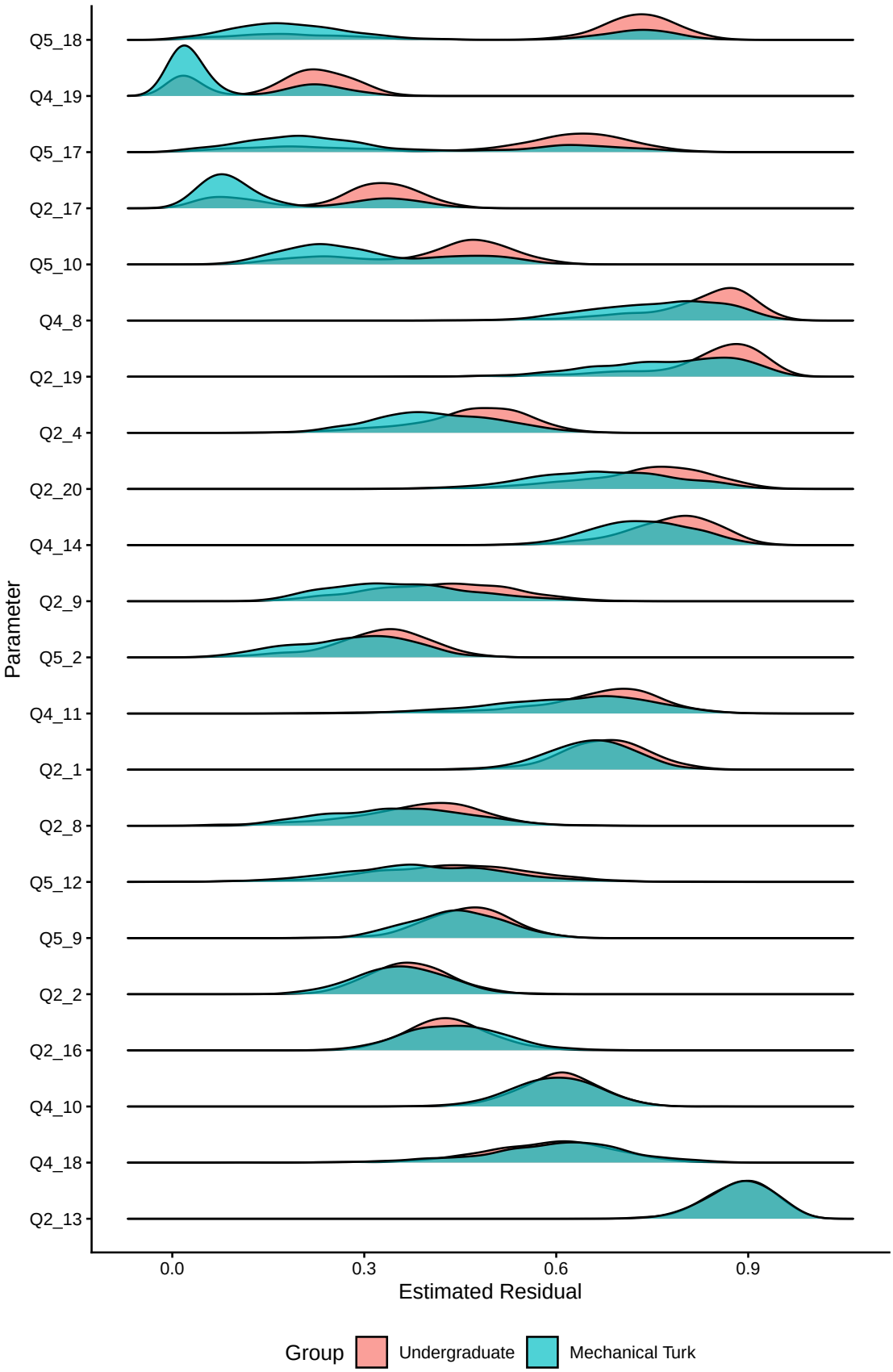

Figure 6. Residual estimation distributions for items that were bootstrapped with original group labels ordered by effect size for non-invariant models.